

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: CRISISLTLSUM | Context: Ecuador — Humanitarian Crisis Social Media Analysts

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Ground-truth labels were produced by three MTurk workers per timeline **[Q56, Q70]**, explicitly restricted to 'countries where native English speakers are most likely to be found' **[Q61]**. The two reference summaries used for ROUGE scoring were chosen as those written by the two most-agreeing workers **[Q87, Q88]**. Summarization quality ratings came from a separate pool of five MTurk workers under the same location restriction **[Q105, Q106]**. No Spanish-speaking, Latin American, or humanitarian-practitioner annotators participated. The Ecuador deployment requires inter-operational validation by teams of Ecuadorian field coordinators, social analysts, and domain experts — explicitly not a crowd-sourced English-speaker pool (elicitation A3, OC priority HIGH). Inter-annotator agreement of 90.06% **[Q77]** and 91.77% against experts **[Q79]** is internally high but provides no evidence about Ecuadorian-stakeholder agreement. This is a direct convergent-validity violation: the labels reflect a different population's judgments than the deployment requires.

ID	Evidence
Q56	"Each HIT contains three noisy timelines, and we collect annotations from 3 different workers on each." (p.5)
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)
Q70	"To make the ground-truth on whether each tweet is part of the timeline or not, we take the majority label among the three workers." (p.6)
Q77	"The average timeline-level agreement between those two workers is 90.06%." (p.6)
Q79	"The average timeline-level agreement rate of this analysis is 91.77%." (p.6)
Q87	"Since each timeline in our dataset is annotated by three workers (§3.2), it includes three summaries." (p.7)
Q88	"To adapt the timeline summarization task to this setting, we pick the two summaries written by the two workers who agree the most based on the timeline extraction labels they assign to the tweets in each timeline." (p.7)
Q105	"Following a similar process in 3.2, we recruit a group of workers from MTurk with the same Location restriction (QC1), who can pass our pilot study as a qualification test." (p.8)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)

Strengths

- The benchmark documents annotator quality control thoroughly (QC1–QC4) **[Q60, Q61, Q62, Q64, Q66]** and reports both inter-worker and worker-vs-expert agreement **[Q77, Q79]**, making the methodology auditable and providing a clear template that an Ecuador-specific re-annotation effort could replicate using inter-operational teams.

ID	Evidence
Q60	"Following prior quality control practices (Briakou et al., 2021), we use multiple quality control (QC) steps to ensure the recruitment of high-quality annotators." (p.5)
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)
Q62	"Next, we recruit annotators who pass our qualification test (QC2), where we ask them to annotate 3 timelines." (p.5)
Q64	"We check annotators' performance on timeline extraction task against experts' labels and have experts manually review (QC3) annotators' summary quality." (p.5)

Q66	"During the final task, we perform regular quality checks (QC4), and only use workers who consistently perform well." (p.5)
Q77	"The average timeline-level agreement between those two workers is 90.06%." (p.6)
Q79	"The average timeline-level agreement rate of this analysis is 91.77%." (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — labels reflect English-speaking MTurk worker judgments [Q61], not Ecuadorian humanitarian-coordinator judgments. The elicitation explicitly identifies an inter-operational team as the authoritative judge.

OC-2: Disagreement is likely substantial: humanitarian coordinators would weight informal community broadcasts, infrastructure-access signals, and displacement indicators more heavily than US MTurk workers, who applied a generic fact-vs-opinion rule [Q58].

OC-3: The benchmark documents annotator location restriction (English-speaking countries) [Q61], qualification testing [Q62], and quality monitoring [Q66], but provides no demographic breakdown beyond that. There is no Datasheet-level disclosure of country, professional background, or domain expertise.

OC-4: Re-annotation by a representative Ecuadorian inter-operational team is required for deployment; the elicitation flags this as a HIGH priority gap.

OC-5: Aggregation by majority vote across three workers [Q70] could erase minority perspectives; for humanitarian content where a single field-relevant signal in a tweet may be operationally critical, majority voting risks suppressing exactly the rare signals coordinators most need.

OC-6: Documented: convergent-validity violation (labels do not correlate with deployment-population judgments) and external-validity violation (US-MTurk judgments do not generalize to humanitarian-coordinator context).

ID	Evidence
Q58	"A tweet is labeled as part of the timeline only if it satisfies all the following three conditions: relevant: talks about the same event indicated in the seed tweet, informative: provides facts about the event but not only contains personal points of view, not repetitive: brings in new information." (p.5)
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)
Q62	"Next, we recruit annotators who pass our qualification test (QC2), where we ask them to annotate 3 timelines." (p.5)
Q66	"During the final task, we perform regular quality checks (QC4), and only use workers who consistently perform well." (p.5)
Q70	"To make the ground-truth on whether each tweet is part of the timeline or not, we take the majority label among the three workers." (p.6)
Q105	"Following a similar process in 3.2, we recruit a group of workers from MTurk with the same Location restriction (QC1), who can pass our pilot study as a qualification test." (p.8)

Information Gaps

- No documented annotator demographics beyond country-of-residence proxy; no comparison of MTurk worker labels against humanitarian-practitioner labels on any subset.

Requires Expert Verification

- The actual labeling protocol used by Cruz Roja Ecuatoriana and OCHA Ecuador for tweet-relevance and summary-quality decisions, and whether existing inter-agency SOPs can be operationalized as an annotation protocol.

Remediation

Gap 1: Ground-truth labels and reference summaries produced by English-speaking-country MTurk workers [Q61] cannot serve as proxies for inter-operational humanitarian-team judgments required by the deployment.

Recommendation: Re-annotate a representative subset using inter-operational teams of Ecuadorian field coordinators, social analysts, and domain experts; replicate the documented QC1–QC4 protocol [Q60–Q66] but adapted for humanitarian practitioners; report agreement separately for each role to surface minority operational perspectives that majority voting would erase.

ID	Evidence
Q60	“Following prior quality control practices (Briakou et al., 2021), we use multiple quality control (QC) steps to ensure the recruitment of high-quality annotators.” (p.5)
Q61	“First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found.” (p.5)
Q62	“Next, we recruit annotators who pass our qualification test (QC2), where we ask them to annotate 3 timelines.” (p.5)
Q63	“We run several small pilot tasks, each with a replication factor of nine.” (p.5)
Q64	“We check annotators' performance on timeline extraction task against experts' labels and have experts manually review (QC3) annotators' summary quality.” (p.5)
Q65	“Only workers passing all the quality control steps contribute to the final task.” (p.5)
Q66	“During the final task, we perform regular quality checks (QC4), and only use workers who consistently perform well.” (p.5)